L3B: Curves and Sound

CS1101S: Programming Methodology

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28th August, 2026

Version 1.0 — Built: 2026-08-28 01:56:37

Outline

Curves

Sound

Mastery Checks

- Oral tests
- Conducted in 3-way meetings:
  - 2 students, 1 Avenger, mostly of the same Studio
- Anytime during the semester
- Topics for Mastery Check 1 (3% of total course grade)
  - Scope
  - Higher-order functions
  - Substitution model and iterative/recursive processes
- Pass/fail for each topic
  - Repetition allowed per topic

Mastery Checks: The Process

- Form pairs. Your Avenger will help you find a partner from a different Studio if necessary
- Make appointment with Avenger
- For each topic, you have 10 minutes to convince the Avenger that both of you have mastered the topic
- Use proper English and terminology; illustrate your argument with examples
- Be ready for using examples given by the Avenger
- Be ready to answer questions on the topic posed by Avenger
- Avenger will pass you on a topic only if s/he is convinced you've mastered it
- The pair can have repeat attempt(s) with the Avenger on unsuccessful topic(s)
- Other staff may check as well for repeated attempts

Try to finish MC1 before Recess Week (great prep for Midterm!)

Logical Operators

- `not p`: true when `p` is false
- `p and q`: true when both `p` and `q` are true
- `p or q`: true when at least one of `p`, `q` is true

`and`/`or` are syntactic forms, not operators

Unlike a real binary operator, the right-hand expression is not always evaluated (SICPy 1.1.6):

`p and q` ≡ `q if p else False`

`p or q` ≡ `True if p else q`

Logical Operators: Laziness

```
def boom():
    return 1 / 0  # would crash if ever evaluated

print(False and boom())
print(True or boom())
```

Neither call to `boom()` ever executes — `and`/`or` stop as soon as the result is decided.

Outline

Path, Missions & Quests on Curves

- Path: “Curves”
- Mission: “Curve Introduction”
- Mission: “Curve Manipulation”
- Mission: “Beyond the First Dimension”
- Quest: “Cardioid Arrest”
- Quest: “Curvaceous Wizardry”
- Contest: “The Choreographer”

Representation of Curves — Explicit Form

- Curves in 2D
  The value of the dependent variable is given in terms of the independent variable $y = f(x)$
  Example: $y = mx + h$ (straight line)
- Some curves cannot be expressed in explicit form
  Examples: vertical straight line, circle
- Curves in 3D
  Requires two equations
  - $y = f(x)$
  - $z = g(x)$

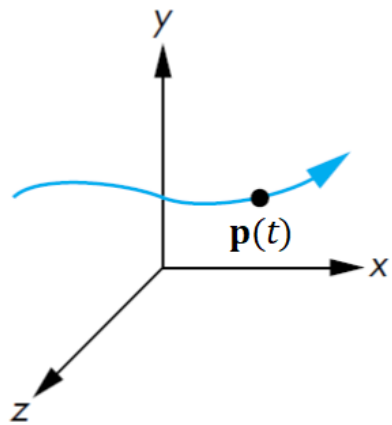
Representation of Curves — Implicit Form

- Curves in 2D
  $f(x, y) = 0$
  Examples:
  - $ax + by + c = 0$ (straight line)
  - $x^2 + y^2 - r^2 = 0$ (circle)
- Curves in 3D
  Can be represented as the intersection of two surfaces
  - $f(x, y, z) = 0$
  - $g(x, y, z) = 0$
- A drawback: Difficult to obtain points on the curves
  Because the equations are just membership tests

Representation of Curves — Parametric Form

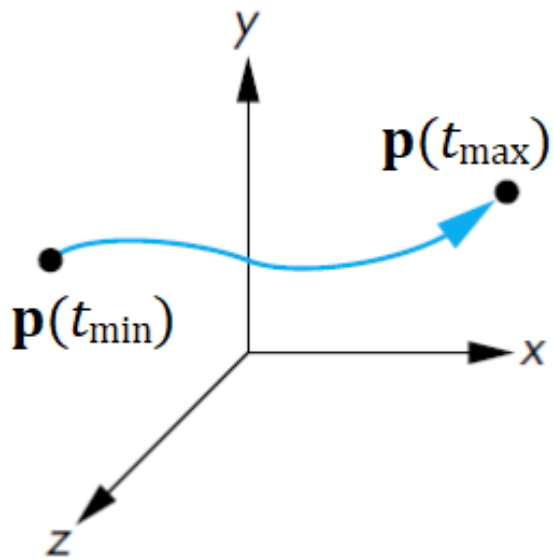
- Curves in 2D and 3D
  Each spatial variable for points on the curve is expressed in terms of an independent variable t, the parameter

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \qquad \mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$



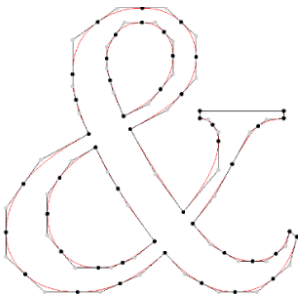
Representation of Curves — Parametric Form (cont'd)

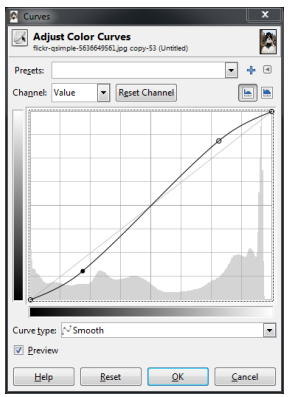
- A curve segment is defined for $t_{\text{min}} \leq t \leq t_{\text{max}}$
  Often, $0 \leq t \leq 1$

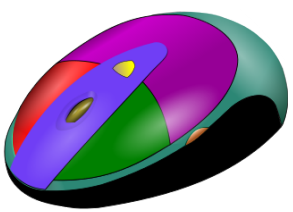


Applications of Curves

- Drawing smooth, vector-based curves at any resolution
- Font design and rendering
- Smooth animation paths (objects, lights, cameras)
- Designing smooth functions
  - E.g. for image color & tone adjustment
- 3D model design and rendering
  - Engineering and industrial design, movies, and games
- Data fitting







Curves in Python

- Supported by the curve module `https://source-academy.github.io/modules/documentation/modules/curve.html`
- Uses parametric representation
- Parameter t is within the unit interval [0, 1]
  - If C is a Curve, its
    - starting point is C(0)
    - ending point is C(1)

Specifications of Curves in Python

Curve

Curve := Number → Point

A Curve is a **function** that takes a **Number** as argument and returns a **Point**. The Number argument must be within the interval [0, 1].

Point

`make_point` : (Number, Number) → Point

`x_of`, `y_of` : Point → Number

`x_of(make_point(n, m))` = n

`y_of(make_point(n, m))` = m

Defining Curves

- Example:

```
def diagonal_line(t):
    return make_point(t, t)

draw_connected(10)(diagonal_line)
```

What does draw_connected do?

- Example:

```
def print_point(p):
    print("point:", x_of(p), y_of(p))

def diagonal_line(t):
    p = make_point(t, t)
    print_point(p)
    return p

draw_connected(10)(diagonal_line)
```

Let's make a general line!

- First Try

```
def line_with(m, h, t):
    return make_point(t, m * t + h)

draw_connected(10)(line_with)
```

Attempt 2:

- Second Try

```
def line_with(m, h):
    return lambda t: make_point(t, m * t + h)

draw_connected(10)(line_with(2, 0))
```

Off the map!

- Second Try

```
def line_with(m, h):
    return lambda t: make_point(t, m * t + h)

draw_connected(10)(line_with(2, 3))
```

A more complex example:

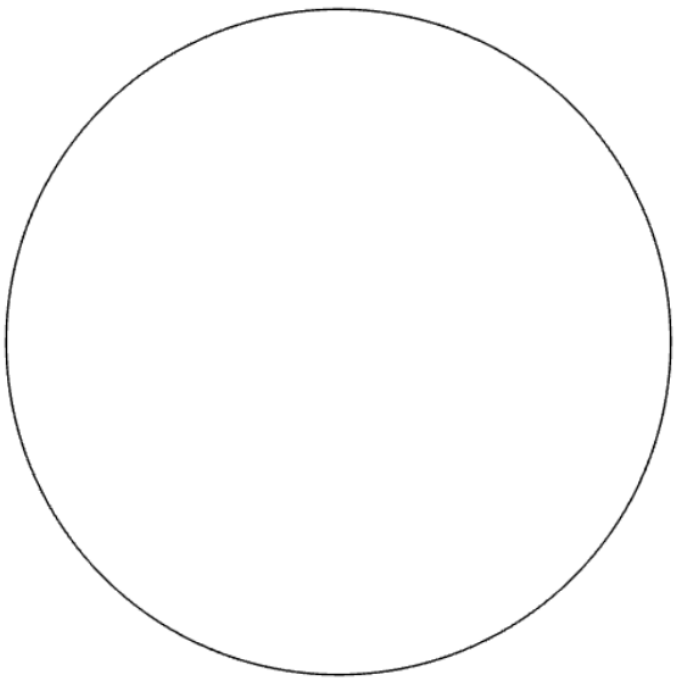
- Example:

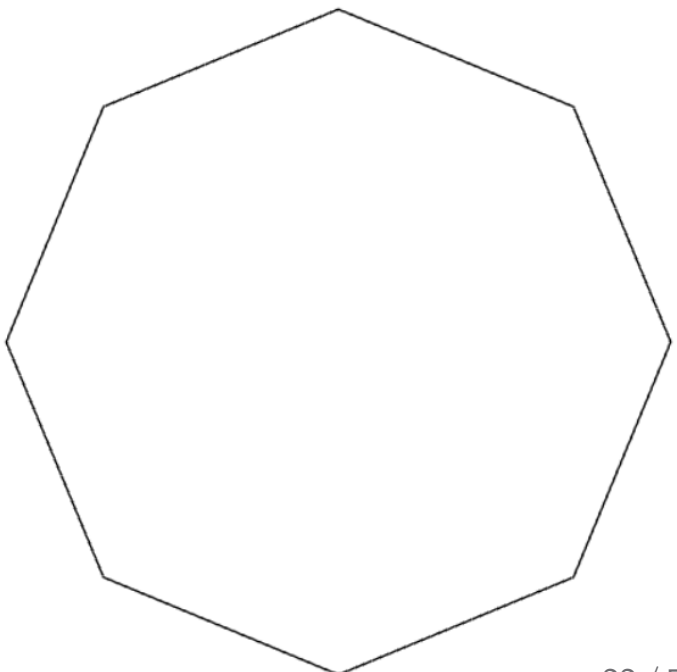
```
def unit_circle(t):
    return make_point(math_cos(2 * math_pi * t),
                      math_sin(2 * math_pi * t))
```

Drawing Curves

- Examples:

```
draw_connected(200)(unit_circle)
draw_connected_full_view_proportional(200)(unit_circle)
draw_connected_full_view_proportional(8)(unit_circle)
```



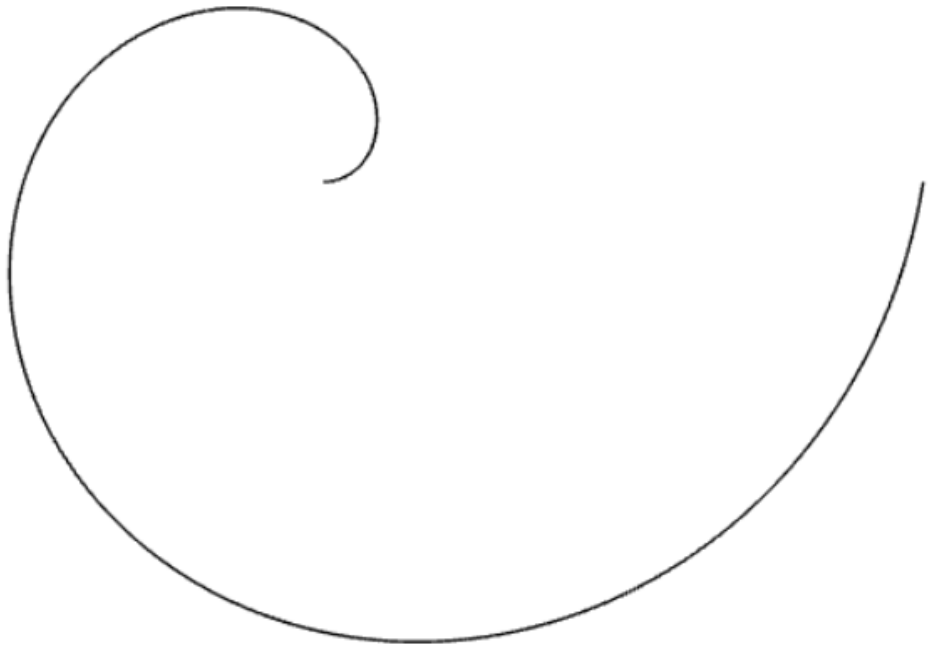


Example: Spiral Curves

- Wanted: Write a function spiral_one that represents a spiral curve
  - Uses unit_circle
  - One revolution only
- Solution:

```
def spiral_one(t):
    p = unit_circle(t)
    return make_point(t * x_of(p), t * y_of(p))

draw_connected_full_view_proportional(200)(spiral_one)
```

Example: Spiral Curves (Result)



Example: Spiral Curves

- Wanted: Write a function spiral that returns a spiral curve
  - Uses unit_circle
  - Number of revolutions is a parameter
- Attempt #1:

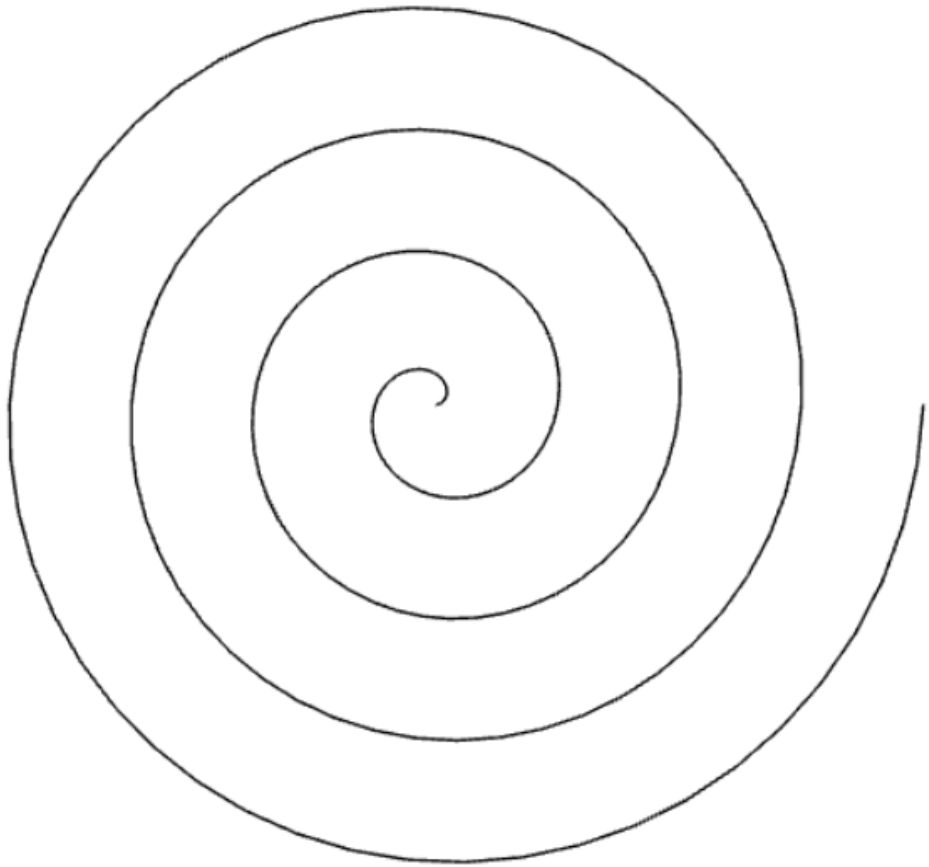
```
def spiral(rev, t):
    p = unit_circle((t * rev) % 1)
    return make_point(t * x_of(p), t * y_of(p))

draw_connected_full_view_proportional(200)(spiral)
```

Example: Spiral Curves

- Wanted: Write a function spiral that returns a spiral curve
  - Uses unit_circle
  - Number of revolutions is a parameter
- Attempt #2:

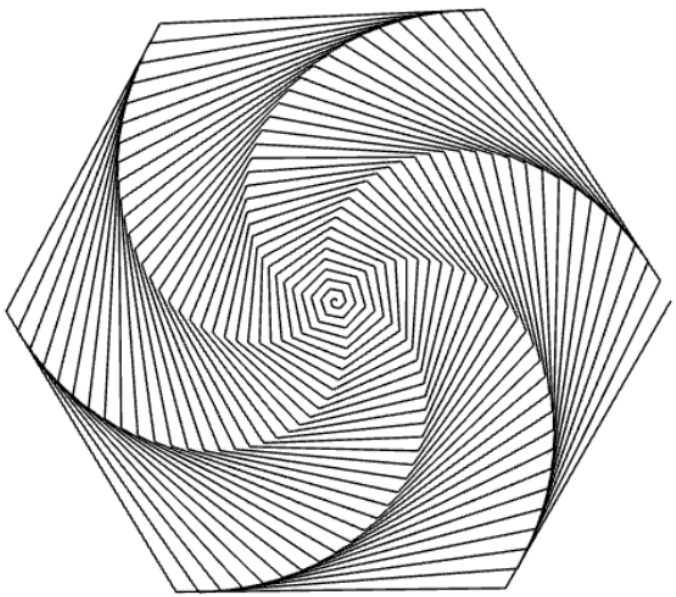
```
def spiral(rev):
    def spiral_helper(t):
        p = unit_circle((t * rev) % 1)
        return make_point(t * x_of(p), t * y_of(p))
    return spiral_helper

draw_connected_full_view_proportional(200)(spiral(4))
```

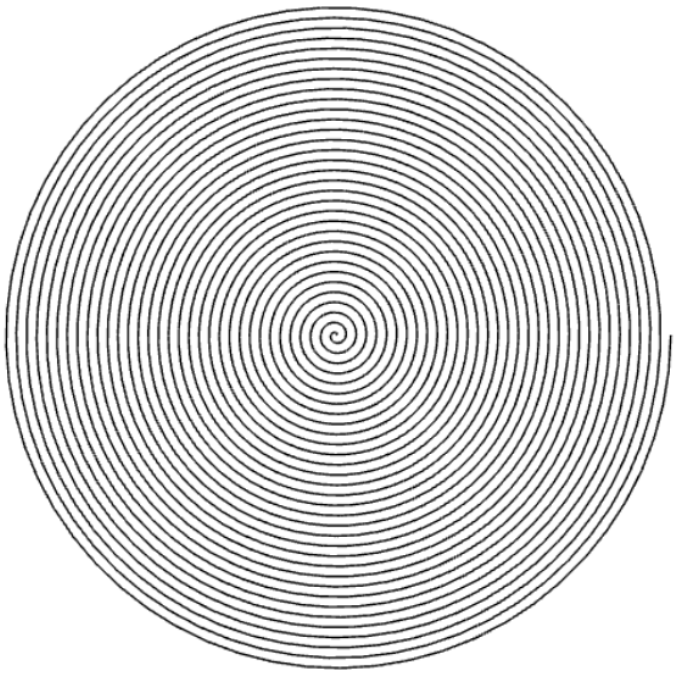
Example: Spiral Curves (Result)



Example: Spiral Curves

```
draw_connected_full_view_proportional(200)(spiral(33))
draw_connected_full_view_proportional(2000)(spiral(33))
```

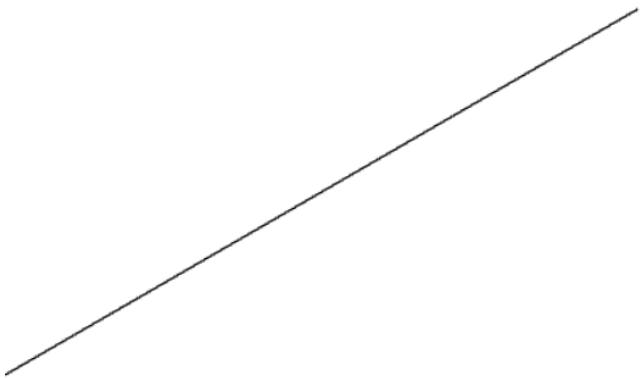




Transformations on Curves

- Rotate `unit_line` by $\pi/6$, then translate it upwards by 0.25.

```
rot_line = rotate_around_origin(math_pi / 6)(unit_line)
shifted_rot_line = translate(0, 0.25, 0)(rot_line)
draw_connected(200)(shifted_rot_line)
```

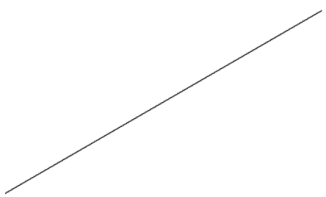


Transformations on Curves

- Alternatively, compose the transformations before applying them to the curve.

```
def compose(f, g):
    return lambda x: f(g(x))
```

```
shift_rot = compose(translate(0, 0.25, 0),
    rotate_around_origin(math_pi / 6))
shifted_rot_line = shift_rot(unit_line)
draw_connected(200)(shifted_rot_line)
```

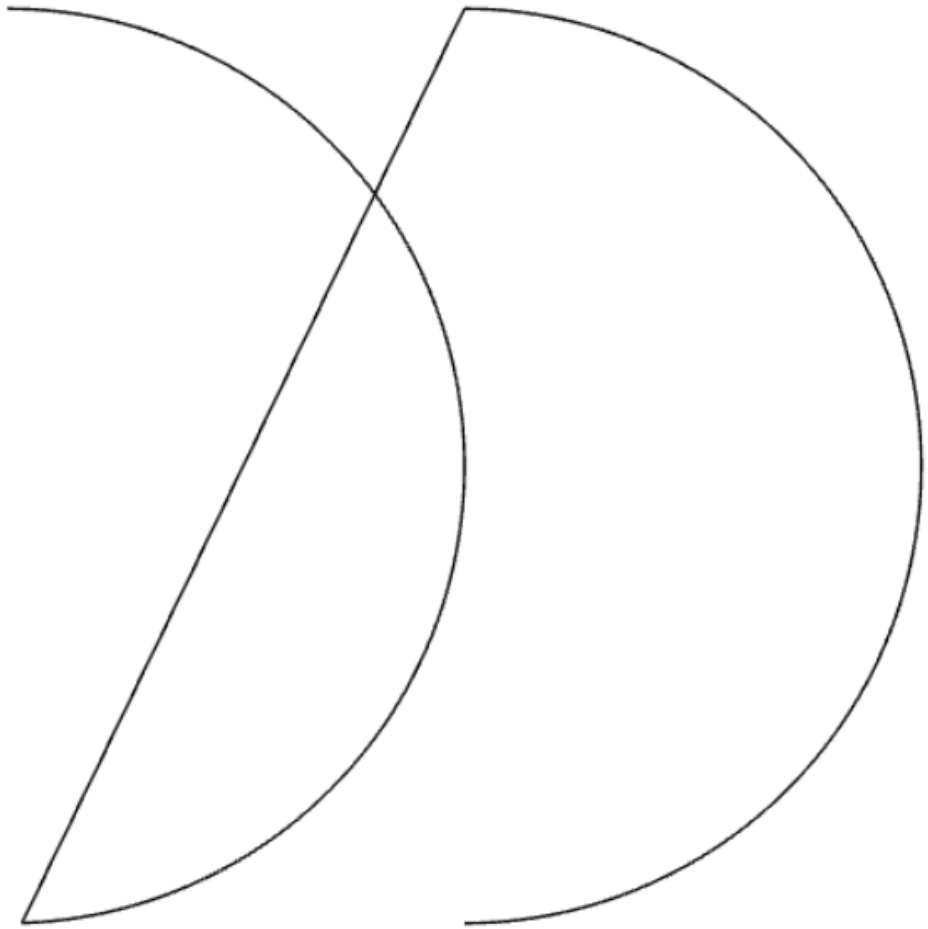


Connecting Curves

- Example:

```
def connect_rigidly(curve1, curve2):
    return lambda t: (
        curve1(2 * t)
        if t < 1 / 2
        else curve2(2 * t - 1)
    )

result_curve = connect_rigidly(arc, translate(1, 0, 0)(arc))
```

Connecting Curves (Result)

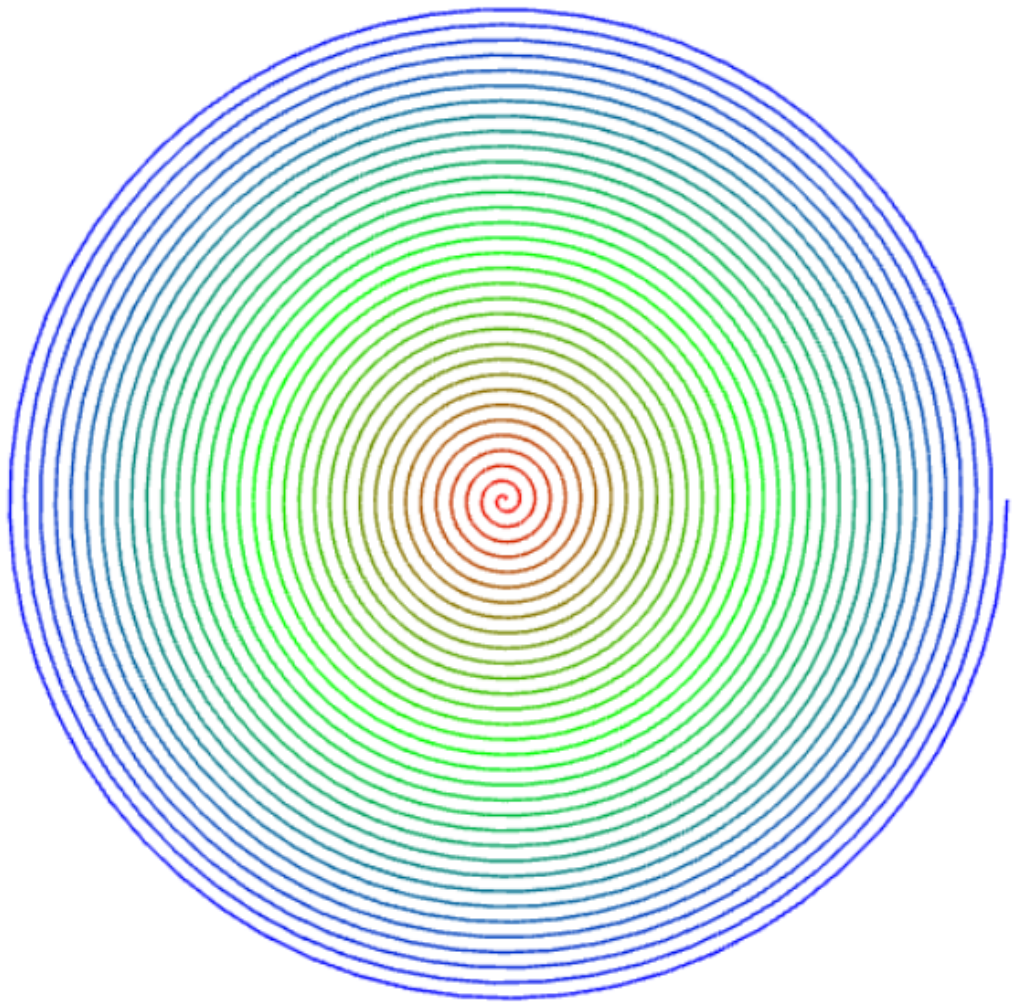


Colored Curves

- Example:

```
def colorful_spiral(rev):
    def spiral_helper(t):
        p = unit_circle((t * rev) % 1)
        R = max(0, 1 - 2 * t) * 255
        G = (1 - abs(1 - 2 * t)) * 255
        B = max(0, 2 * t - 1) * 255
        return make_color_point(t * x_of(p),
            t * y_of(p),
        R, G, B)
    return spiral_helper
```

Colored Curves (Result)

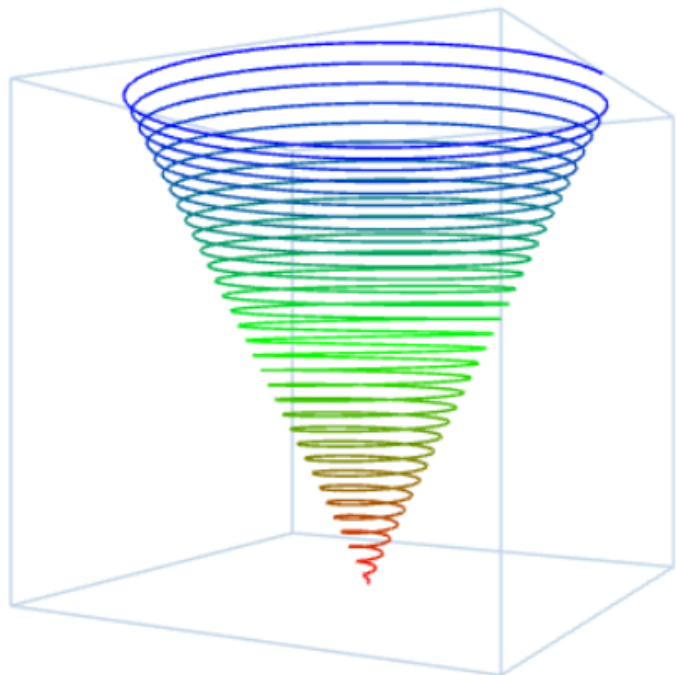


3D Curves

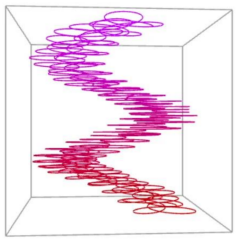
- Example:

```
def colorful_3D_spiral(rev):
    def spiral_helper(t):
        p = unit_circle((t * rev) % 1)
        R = max(0, 1 - 2 * t) * 255
        G = (1 - abs(1 - 2 * t)) * 255
        B = max(0, 2 * t - 1) * 255
        return make_3D_color_point(
        t * x_of(p), t * y_of(p), 2 * t, R, G, B)
    return spiral_helper
```

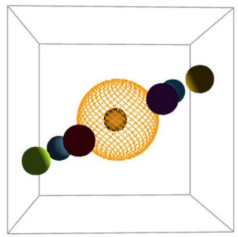
3D Curves



Curves from Past Contests



Link to code



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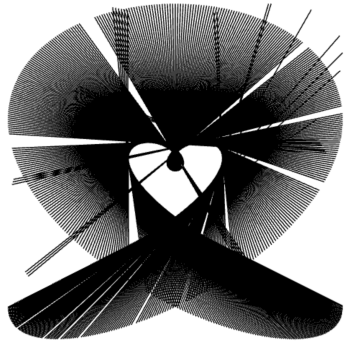

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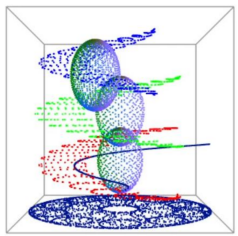

Link to code

Outline

Missions & Quests on Sound

- Mission: “Premorseal Communications”
- Mission: “POTS and Pans”
- Mission: “Musical Diversions”
- Quest: “Echoes of the Past”
- Quest: “The Magical Tone Matrix”
- Contest: “Game of Tones”

Sound

- What is Sound?
- Digital Sound
- Specifications of Sound in Python

Specifications of Sound in Python

Wave

Wave := Number $\rightarrow$ Number

A Wave is a **function** that takes a **Number** (time, in seconds) as argument and returns a **Number** (amplitude, between -1 and 1).

Sound

`make_sound` : (Wave, Number) $\rightarrow$ Sound

A Simple Example

- Simple example:

```
pitch_A_wave = lambda t: math_sin(2 * math_pi * 440 * t)
pitch_A = make_sound(pitch_A_wave, 1.5)
play(pitch_A)
```

How does make_sound work?

- Simple example:

```
def pitch_A_wave(t):
    print("current t:", t)
    return math_sin(2 * math_pi * 440 * t)

pitch_A = make_sound(pitch_A_wave, 0.1)
play(pitch_A)
```

C Major Chord

- Second example:

```
C_maj_chord_wave = lambda t: (
    0.33 * math_sin(2 * math_pi * 261.63 * t) +
    0.33 * math_sin(2 * math_pi * 329.63 * t) +
    0.33 * math_sin(2 * math_pi * 392.00 * t)
)
C_maj_chord = make_sound(C_maj_chord_wave, 1.5)
play(C_maj_chord)
```

Do, a deer, a female deer

- Third example:

```
doremi_wave = lambda t: (
    math_sin(2 * math_pi * 261.63 * t)
    if t < 0.5
    else math_sin(2 * math_pi * 293.66 * t)
    if t < 1.0
    else math_sin(2 * math_pi * 329.63 * t)
)
doremi = make_sound(doremi_wave, 1.5)
play(doremi)
```

Sound in Python

- Supported by the sound module `https://source-academy.github.io/modules/documentation/modules/sound.html`

Uses functional sound!

Sample Songs

Mission Impossible

```
play_in_tab(mission)
```

Minecraft Theme

```
play_in_tab(minecraft_theme)
```

Testing... Testing... 1, 2, 3...

Time for a Sound Check(-In)!

Look for Check-In I3B in the Source Academy!

Summary

- We can create curves by defining a function that maps from a parameter to a Point.
- The curve is drawn by sampling our function and generating points and interpolating between them.
- We can also create sounds by defining a function that describes the sound over time.
- The audio is synthesized by sampling our function and generating points and then interpolating between them.
- We saw the sound and curve modules in the Source Academy!
- There are contests coming up that use these!

Unit 1 is officially over wooohooo! (except for S4 and RA1)